Econometric Game 2026
Case A

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Center for Research in Energy
ECONOMICS AND MARKETS

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1 Instructions for structuring the report

Your report should be clear, structured, and well-argued, demonstrating both methodological rigor and economic intuition. The report should not exceed 25 pages including Appendix. Below is the recommended structure:

1. Introduction

   Briefly introduce the problem and its economic relevance. State your research questions and objectives. Outline the structure of the report.

2. Data Description

   Provide relevant descriptive statistics for selected variables in the dataset that you consider particularly relevant for the analysis. What are the main features of the different time series, such as volatility and price span, periodicity, seasonality, etc.? Explain key variables, transformations, and any data cleaning steps. Highlight with appropriate graphs and tables.

3. Methodology

   Clearly explain the econometric model(s) applied. Justify the choice of estimation and forecasting techniques. Discuss any assumptions and potential econometric issues.

4. Empirical Results

   Present key estimation and forecasting results in tables and figures and conduct appropriate interpretations. Perform robustness checks and discuss potential limitations.

5. Discussion and Economic Interpretation

   Relate and interpret your findings for real-world implications and refer where relevant to economic theory including implications for electricity trading over various horizons. Discuss any policy recommendations or business implications if relevant.

6. Conclusion

   Summarize key findings concisely. Highlight limitations and potential extensions for future research.

7. References

   List all academic papers, datasets, and other sources cited in a consistent format. If using ChatGPT or similar as a reference source, make the appropriate reference and possibly with quotation marks.

8. Statement about the use of hardware and software

   There are no restrictions on the use of hardware or software. However, use of cloud computing resources are not permitted. The report should include a statement about **all** the computing software that has been used to work with the Case. Note particular guidelines about using ChatGPT and other AI tools.

9. Statement about use of ChatGPT

   Compulsory: In your report, your reference list should include a statement explaining exactly how ChatGPT and other AI tools have been used in working with your report. If the statement is absent, the report will be withdrawn from assessment. See also section 3 below.

10. Appendix (if necessary)

    Include additional tables, robustness tests, or technical derivations that support the main text.

2 Instructions for submitting the report

- You can send any questions about the case or the competition to econometricgame@vsae.nl
- You are not allowed to work on the report outside the competition hours. That is why we ask you to submit the (working) report on Wednesday before 18:00 by sending it to juryeg@vsae.nl. Subject of this email: "First Day Case A - Team [Number]". Use the number that was communicated to you in the email.
- The deadline for the final report submission is on Thursday at 17:00. You should do so by sending it to juryeg@vsae.nl. Please put your team number on the title page of your report. Do not include your names and your university name. Subject of this email: "Final Case A - Team [Number]".
- Please note, the total volume of the report including Appendix should not exceed 25 pages. Be mindful of page formatting: use standard font size (e.g., 12 pt), appropriate margins (e.g., 1 inch), and spacing (e.g., 1.5 or 2x line spacing).

3 Guidelines for using ChatGPT and other AI tools

- Allowed Usage:
  ChatGPT can be used for instance to assist with coding tasks (including debugging), optimization, and understanding programming concepts. You may use ChatGPT to generate code snippets, explore different coding approaches, and troubleshoot errors.
- Not Allowed Usage:
  ChatGPT cannot be used to *write* your report, including explanations, analysis, or conclusions. All written content in your report must be your own work, based on your understanding and research. If you use ChatGPT as a source for your report, you should indicate this explicitly by appropriate referencing like any other references.
- **Compulsory: In your report, your reference list should include a statement explaining exactly how ChatGPT and other AI tools have been used in working with your report. If the statement is absent, the report will be withdrawn from assessment.**

4 Case A

Trading in energy markets is rather unique. The electricity market is especially interesting because electricity has several unusual characteristics: it depends heavily on production from renewable sources such as wind and solar, the transmission infrastructure grid, and it is a commodity that cannot be easily stored. Since the beginning of the 21st century, the liberalization of Europe's electricity markets has created many new opportunities for energy producers and trading firms, much of it taking place on the so-called day-ahead market. In 2024, trading volumes on the EPEX spot markets reached 868 TWh (EPEX SPOT SE, 2025).

The spot market—often called the day-ahead market—sets the electricity price for each hour of the next day through an auction that matches expected supply with anticipated demand. Importantly, this means that, for example, prices of hours 5 and 6 of the next day are determined simultaneously[1]. In other words, price for one hour cannot be used to predict the next hour and only information available the day before can be used for such a prediction. Throughout the rest of the case description we will refer to these day-ahead prices as spot prices. In electricity trading, market participants can make agreements at a range of horizons: day-ahead, week-ahead, month-ahead, or even year-ahead, often by trading futures based on day-ahead spot prices. For traders, producing reliable price forecasts at different time horizons is essential. A trader might need a forecast up to a year into the future, using only the information available today. A good forecast can lead to significantly higher profits, while a poor one can result in substantial losses.

You are provided with several datasets containing key variables for the analysis of electricity prices, ranging from 01/01/2018 to 31/12/2023 for Denmark, price zone DK1 (West Denmark). In particular the dataset will contain,

- Electricity Prices: We provide hourly spot electricity prices, coming from day-ahead auctions. Recall that these prices are peculiar, in the sense that the prices for all the hours within the following day are determined jointly at noon of the auction day. This may be relevant for the analysis.
- Electricity Loads: We will provide the data for hourly loads, both their realization and their day-ahead forecast. These quantities represent the demand of electricity and can affect the price.
- Renewable Sources Production: We provide hourly data for production of renewable sources, wind and solar, both their realization and day ahead forecast. The price of electricity is heavily affected by the production of renewables, since when wind and solar production decreases, the demand requires the intervention of more expensive sources such as fossil fuels to balance markets.

[1] Have a look at the Nordpool Group and Energinet websites for more insights into the production of electricity and the transmission network.

- Energy: We provide daily prices for Natural Gas, Oil, Coal and European Union carbon emission allowances (EUA). For all these series, we provide the open and close price, the lowest and highest value that the price has reached in a day and the % change with respect to the previous day.
- Macro-Financial: We provide mixed frequency data concerning a number of different macroeconomic and financial indicators. In particular, the dataset includes the daily series of the VIX, EU STOXX 50 and MSCI EU indexes and the Euro/USD exchange rate. The dataset includes also monthly data on Danish total and energy specific CPI. Finally, we provide quarterly Danish and Euroarea GDP.

With this dataset you are required to address the following points:

- Provide relevant descriptive statistics for selected variables in the dataset that you consider particularly relevant. What are the main features of the different time series, such as volatility and price span, periodicity, seasonality, etc.? Highlight your findings in the report with appropriate graphs and tables.
- The primary goal is to produce forecasts for the hourly electricity prices for the year 2024. We can divide this into two tasks:
    1. First, you are asked to produce forecasts of the spot prices for all hours for each day of the first week of the year, using only data available up to the 31st December 2023. This amounts to 7×24 forecasts.
    2. Second, you are asked to produce forecasts of the weekly average of hourly spot prices for every week of 2024, using again only data available up to the 31st December 2023. In other words, the variables of interest are the weekly averages of each hourly price. This amounts to 52×24 forecasts.

    You will also be given the observed electricity prices in 2024, to compare and evaluate your forecasts with the actual observed prices.
- Given the high amount of forecasts you have to produce, what is the right approach to evaluate these forecasts, and are they all equally relevant? Report these forecast evaluations for your chosen model and discuss the results.

    Hint: Recall that you can trade futures at shorter horizons too. Hence, it might be relevant to consider subsets of the forecast for evaluation.

5 Data Description

Electricity Prices

This folder contains the target variable for Case A and several explanatory variables, such as the day-ahead forecast of Loads, Wind and Solar production and the exchange between different price zones. The files contain several forecasts and the timing convention is the following. The forecasts are day-ahead and place in the entry they forecast. In other words, the forecast column for a datapoint in the entry at day d and hour h was available on day $d-1$ and can be used in the model that forecasts the price at day d and hour h.

- **Prices:** This folder contains the spot prices. In particular every, *csv* file contains the prices for a whole calendar year. The Date column ("MTU (UTC)") identifies the day and hour of the price while the "Day-Ahead Prices" contains the main variable. You can ignore empty columns or identifiers as the price zone or the sequence column.
- **Loads:** This folder contains the electricity Loads. The Date column ("MTU (CET/CEST)") identifies the day and hour. The "Actual Total Load (MW)" identifies the Load for that date, the "Day-ahead Total Load Forecast (MW)" is a forecast made the previous day for the current entry.
- **Renewables:** This folder contains the forecast of the production of electricity from renewable sources. The files are divided into solar, and onshore and offshore wind. Each file contains the dates, the actual production and the day-ahead forecast of the production. The current and intra-day columns contain different estimates of the actual quantity elaborated at different time intervals. These cannot be used at day $d-1$ as they were not available.
- **Transmission:** This folder contains the electricity flow between price zones. In particular for every date the flow between price zones is reported together with the in and out zones.

Energy Series

This folder contains the prices for several key resources for energy market, such as Coal, Oil, Natural Gas and Carbon emission allowances. The files contain daily observations and the columns have the following naming convention:

- Price : close price for the reference series, intended as main variable.
- Open : price of the given series at the opening of the trading day.
- High : highest value of price during the trading day.
- Low: lowest value of price during the trading day.

- Volume: volume of trading in the given day.
- Change: percentage change over the day.

Macro Finance

This folder contains several key macroeconomic and financial indicators for the economic area of Denmark. In paricular, we divide between,

- Daily Data: This consists of the VIX, the MSCI and EuroSTOXX50 indexes and the EU/USD exchange rate. Some of these variables report both open and close prices and information on the trading day as the energy series, some report only the daily index.
- Monthly Data: This consists of Danish general and energy specific CPI.
- Quarterly Data: This consists of Danish and EuroArea GDP.

Climate data

This folder contains daily weather observations from the *European Climate Assessment & Dataset (ECA&D)*. Observations are provided at the station level and then *assigned* to electricity bidding zone DK1 using station coordinates. Each file contains one row per station-day (i.e., at most one observation per station and calendar day in the final datasets). Only observations flagged as *valid* by ECA&D quality control are retained; consequently, some stations have missing days.[2]

For each climate variable: `tg` (mean temperature), `rr` (precipitation), `ss` (sunshine duration), and `fg` (mean wind speed) there is (`.csv.gz`) file.

Columns (shared across all climate files). All climate files share the following columns:

- `DATE`: Observation date (daily frequency; formatted as `YYYY-MM-DD` in the exported files).
- `staid`: ECA&D station identifier.
- `station_name`: Station name.
- `ZONE_ID`: Electricity bidding zone to which the station is assigned.
- `LAT_DEC`, `LON_DEC`: Station latitude/longitude in decimal degrees. ECA&D stores coordinates in degrees–minutes–seconds (DMS), e.g. `+55:51:00`, which we convert to decimal degrees using

$$\text{decimal} = \text{sign} \times \left(\text{deg} + \frac{\text{min}}{60} + \frac{\text{sec}}{3600} \right).$$

[2] More information on ECA&D is available at `https://www.ecad.eu`.

- `elevation_m`: Station elevation in meters.
- `ELEID`: ECA&D element identifier describing the measurement/aggregation convention used for the daily value (e.g., the time window and averaging/accumulation rule). The set of possible `ELEID`s differs by variable and may vary across stations (even within stations across dates).

Mean temperature (`tg`)

- **Element identifiers (ELEID) observed in the datasets:**
  - **TG5:** Mean temperature calculated as average of TN and TX (unit: 0.1 °C)
  - **TG23:** Mean temperature 0–0 local time (with changes to winter and summer time) (unit: 0.1 °C)
- **Last column `tg`:** Mean daily temperature (0.1 °C).

Precipitation (`rr`)

- **Element identifiers (ELEID) observed in the datasets (union across zones):**
  - **RR13:** Precipitation amount 22–22 or 23–23 UT (unit: 0.1 mm)
  - **RR19:** Precipitation amount 0–0 local time (with changes to winter and summer time) (unit: 0.1 mm)
- **Last column `rr`:** Daily precipitation amount (0.1 mm).

Sunshine duration (`ss`)

- **Element identifiers (ELEID) observed in the datasets :**
  - **SS6:** Daily sunshine 0–0 local time (with changes to winter and summer time) (unit: 0.1 Hours)
- **Last column `ss`:** Daily sunshine duration (0.1 hours).

Mean wind speed (`fg`)

- **Element identifiers (ELEID) observed in the datasets:**
  - **FG5:** Average wind speed 18–18 UT, 8 three-hourly observations (unit: 0.1 m/s)
  - **FG17:** Average wind speed, unknown interval (unit: 0.1 m/s)
  - **FG19:** Average wind speed 0–0 local time (with changes to winter and summer time) (unit: 0.1 m/s)
- **Last column `fg`:** Mean wind speed (0.1 m/s).

References

EPEX SPOT SE, 2025, "Annual Power Trading Results 2024," working paper, EPEX SPOT SE, Amsterdam, Berlin, Bern, Brussels, London, Paris, Vienna, Press release and annual trading results.